

Lab Task 3: Understanding Single-cast and Multi-cast Delegates

Questions:

1. Define a Delegate:

- Create a delegate `public delegate void MessageHandler();`

2. Create Two Methods:

- Define two methods `ShowWelcome()` and `ShowGoodbye()` that print welcome and goodbye messages.

3. Single-cast Delegate:

- Create a delegate instance pointing only to `ShowWelcome()`.
- Invoke the delegate and observe the result.

4. Multi-cast Delegate:

- Add `ShowGoodbye()` to the same delegate using `+=`.
- Invoke the delegate again and observe both messages are displayed.

5. Remove a Method:

- Use `-=` to remove `ShowWelcome()` from the delegate.
- Invoke the delegate and observe only `ShowGoodbye()` is called.

Lab Task 4: Using Lambda Expressions with Action, Func, and Predicate

Objective:

Practice creating and using lambda expressions with Action, Func, and Predicate delegates in C#.

Tasks:

- 1.** Define an Action<List<string>> delegate using a lambda expression that prints each element of a given list on a new line.
- 2.** Define a Func<int, bool> delegate using a lambda expression that determines whether a given integer is even.
- 3.** Define a Predicate<int> delegate using a lambda expression that checks whether a given integer is a multiple of 5.